

W1 MEASURING GDP
Macro: whole-economy output, prices, jobs, cycles, growth, open-economy links. Time horizons: **SR** demand/rigid prices, **LR** productivity/capital/TFP.
GDP: market value of final g/s produced within borders in period.
 $Y = C + I + G + NX$
 $NX = X - M$. Final only; avoid double counting; inventories count as I .
3 methods: production/value added $\sum(PQ - \text{intermediate})$; expenditure: income $w + r + \pi + \text{rent} + \text{taxes} - \text{subsidies}$. Same circular flow if measured cleanly.
GDP vs GNP: GDP by location; GNP/GNI by residents' income. Foreign firm in Aus raises Aus GDP; profits to foreign owners raise foreign GNP.
Nominal vs real: nominal uses current prices; real uses base-year prices. $GDP_{deflator} = 100 \times NGDP / RGDP$
approx deflator inflation = $\Delta NGDP - \Delta RGDP$.
Pitfalls: used goods no current production; shares/bonds no output; imports subtract only because included in $C/I/G$; govt services valued at cost; home production excluded.

Item	Counts?	Why
new house	I	new capital
used car	No	past output
unsold output	I	inventory
import car	C and -M	not domestic
bank shares	No	asset swap

Mini ex: wheat \$3, flour \$8, bread \$12. $VA=3+5+4=12$; GDP final-good value = 12.

W2 CPI INFLATION SAVING
CPI: fixed-basket Laspeyres index of household consumption prices.
 $CPI_t = 100 \times \frac{\sum p_{it} q_0}{\sum p_{i0} q_0}$
 $\pi_t = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} \times 100$.

Deflating/indexing: real value = nominal/($P/100$); indexed nominal = real target $\times (P/100)$. CPI includes imports; GDP deflator includes domestic output.

Inflation costs: menu, shoe-leather, tax distortions, planning noise, relative-price confusion, unexpected redistribution. Fisher: $r \approx \pi$
ex post $r \approx i - \pi$.

Redistribution: unexpected high inflation helps nominal borrowers, hurts lenders; unexpected deflation does reverse. Anticipated inflation is priced into i .

Saving/wealth: private saving $S_p = Y + NFP - T + TR - C$; public saving $S_g = T - TR - G$; national saving $S = S_p + S_g$. Wealth is stock; saving is flow.

Bias: substitution bias and quality adjustment make CPI overstate cost-of-living; new goods/outlet substitution matter. Core inflation strips volatile items.

Index	Basket	Use
CPI	household fixed basket	cost of living
Deflator	current domestic output	all GDP prices
Core CPI	trimmed/weighted menu	trend inflation

Mini ex: basket $2a + 3b$; base cost 20, current 24 $\Rightarrow CPI = 120$, inflation 20%.

W3 LABOUR CYCLES
Labour force: $LF = E + U$; working age = $LF + NILF$.
 $u = U/LF$
 $part = LF/WAP$, $empratio = E/WAP$. Underemployment is employed but wants more hours.

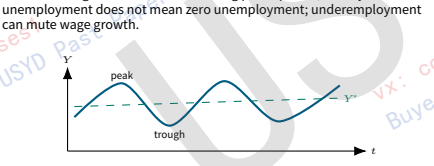
Unemployment: frictional search/matching; structural skills/location/wage rigidity; cyclical demand gap. Natural rate/NAIRU = frictional+structural.

Output gap:
 $gap = \frac{Y - Y^*}{Y^*} \times 100$
Positive gap: $Y > Y^*$, $u < NAIRU$, inflation pressure. Negative gap: spare capacity, disinflation pressure.

Business cycle: peak $\rightarrow$ contraction $\rightarrow$ trough $\rightarrow$ expansion. Recession often 2 quarters negative GDP growth, but broader dating uses output/jobs/income.

Okun: extra cyclical unemployment maps to lost output. Aus rule of thumb: 1 pp cyclical $u \approx 1.8$ pp negative output gap.

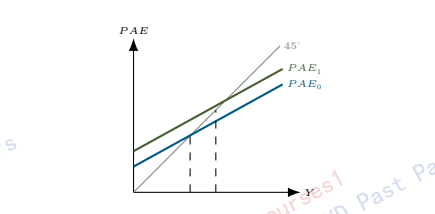
Pitfalls: falling u can coexist with falling participation; zero cyclical unemployment does not mean zero unemployment; underemployment can mute wage growth.



Exam move: classify shock first (AD demand vs SRAS cost vs LRAS capacity), then infer Y , u , P , π .

W4 KEYNESIAN CROSS
Planned aggregate expenditure: $PAE = C + I_p + G + NX$. Equilibrium $Y = PAE$
unplanned inventory $\uparrow$ when $Y > PAE$ so firms cut output.
Consumption: $C = C_0 + c(Y - T + TR)$, $0 < c < 1$, $c = MPC$, $MPS = 1 - c$. In open economy, induced imports reduce domestic multiplier.
Multiplier
2-sector $k = 1/(1 - c)$; with tax/import leakages
 $k = 1/(1 - c(1 - t) + m)$ style. $\Delta Y = k \Delta A$.
Paradox thrift: higher desired saving lowers C_0 , shifts PAE down; SR income falls so actual saving may not rise. LR saving can fund investment, but SR demand falls.

Adjustment: shock ΔA shifts PAE parallel; new intersection with 45-degree line. Larger MPC $\Rightarrow$ steeper PAE and bigger multiplier.



Shock	PAE	Y
confidence up	$C_0 \uparrow$	up
rate up	$I \downarrow$	down
exports up	$NX \uparrow$	up
thrift up	$C_0 \downarrow$	down

Past focus: compute multiplier, compare tax vs G effect, interpret spare capacity/crowding in/out.

W5 FISCAL MONETARY
Fiscal stabilisation: expansionary $G \uparrow$, $T \downarrow$, $TR \uparrow$ closes recessionary gap; contractionary does reverse. Direct G effect exceeds equal tax/transfer effect when $MPC < 1$.
Balanced budget
 $\Delta G = \Delta T$ often raises Y because G enters PAE directly, tax works through c . If tax multiplier = -1.8 , G multiplier = 2, equal \$100m raises Y by \$20m.

Automatic stabilisers: taxes fall and transfers rise in recession without new law; reduce multiplier volatility. Discretionary policy faces recognition/implementation/impact lags.

Budget: deficit = $G + TR - T$; structural at Y^* ; cyclical due to gap. Debt sustainability depends on interest-growth differential and primary balance.

Money: medium exchange, unit account, store value. Money supply = currency + deposits; reserves are bank assets, not household money.

Bank multiplier
desired reserve-deposit ratio r_r , simple deposit multiplier $1/r_r$; if banks hold bonds/extras/reserves, actual expansion smaller.

Tool	Demand effect	Risk
$G \uparrow$	direct AD up	debt/crowding
$T \downarrow$	disp inc up	saved leakage
$TR \uparrow$	targeted AD	inflation if supply bound
stabilisers timely		not enough alone

Equity-efficiency: progressive taxes/transfers reduce inequality but may affect incentives; supply-side public investment can shift LRAS.

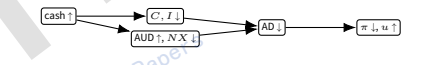
W6 MONETARY POLICY
RBA: targets inflation/stability via cash rate; open market purchase credits reserves, pushes cash rate down; sale drains reserves. Stance judged relative to neutral rate.
Transmission: cash rate $\downarrow \Rightarrow$ market rates $\downarrow \Rightarrow C, I$ up; asset prices up; AUD depreciates; NX up; AD up; unemployment down; inflation pressure up.

Quantity theory
 $MV = PY$; growth form $\mu + v = \pi + g_Y$. If V stable, sustained money growth above real growth creates inflation.

Bonds: price and yield move inversely. Higher market rates reduce price of existing fixed-coupon bonds. Bank solvency: assets-liabilities-equity.

Neutral rate: if cash $< r^*$ policy expansionary even after a small hike; if crosses above r^* , stance contractionary. Zero lower bound can limit cuts.

Tradeoffs: tightening lowers demand/inflation but can appreciate currency and reduce exports; cannot directly fix supply bottlenecks or raise LR capacity.



Exam phrasing: separate target-rate implementation (reserves/cash market) from macro transmission (AD/inflation).

W7 MIDSEM SYNTHESIS
Measurement bundle: output, price level, labour market and rates must be read together. A nominal rise can be price, quantity, or both; deflate before interpreting welfare.
Policy diagnosis: 1 identify gap; 2 classify shock; 3 choose policy; 4 trace channels; 5 mention constraints (lags, ZLB, debt, supply bottleneck, open-economy leakages).

State	Gap	Inflation	Likely response
recession	$Y < Y^*$	low/falling	ease fiscal/monetary
boom	$Y > Y^*$	rising	tighten
stagflation	$Y < Y^*$	high	targeted support+supply
supply boom	$Y > Y^*$	low	avoid over-tightening

Cyclical vs structural budget: at potential, deficit is structural; in recession, observed deficit includes automatic cyclical component.

Common traps: treating financial asset purchases as investment; using population not labour force for u ; saying high inflation always means high real rate; ignoring imports in multiplier.

Compact formula state: $Y = C + I + G + NX$; $u = U/LF$; $gap = (Y - Y^*)/Y^*$; $r \approx i - \pi$; $k = 1/(1 - MPC)$.
Answer template: " This shifts [curve/component] because [mechanism]. In SR [Y,P,u] move. In LR [wages/expectations/capital] adjust if no policy."

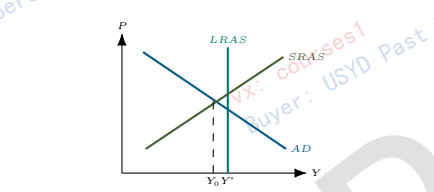
W8 AD-AS POLICY

AD: downward sloping via wealth, interest-rate and exchange-rate channels. Shifts with C, I, G, NX , confidence, taxes/transfers, foreign income, monetary/fiscal policy.

SRAS: upward sloping because input prices/wages sticky. Shifts with nominal wages, imported input costs, productivity, taxes/regulation, supply chains.

LRAS: vertical at Y^* ; shifts with labour, capital, human capital, technology, institutions/resources. Capital stock increase shifts both SRAS and LRAS right.

Self-correction: recessionary gap $\Rightarrow$ wage/cost pressure down $\Rightarrow SRAS$ right; expansionary gap $\Rightarrow$ wages up $\Rightarrow SRAS$ left.



Shock	Curve	SR result
confidence up	AD right	$Y, P \uparrow$
oil price up	SRAS left	$Y, P \uparrow$
tech up	LRAS/SRAS right	$Y^*, P \downarrow$
cash rate up	AD left	$Y, P \downarrow$

Policy mix: demand shock: stabilise AD. Supply shock: output-inflation tradeoff; combine targeted relief, supply repair, cautious monetary policy.

W9 LONG-RUN GROWTH
Living standards: labour productivity Y/L drives average real income; growth sources: capital deepening $K/L \uparrow$, human capital, TFP/technology/institutions, labour utilisation.

Production: $Y = A F(K, L)$; Cobb-Douglas $Y = A K^\alpha L^{1-\alpha}$, so $y = A k^\alpha$

Constant returns to scale: scaling K, L scales Y .

Growth accounting: output per worker growth $\approx$ TFP growth $+\alpha$ growth of K/L . Capital widening keeps K/L constant as L grows; deepening raises K/L .

Policies: education, infrastructure, R&D, property rights, competition, openness, stable macro policy; mere demand stimulus cannot permanently raise Y^* unless it builds capacity.

Diminishing MPK: capital deepening raises y but by less than proportional amount; long-run sustained per-worker growth needs TFP growth.

Mini ex: $Y = 2 K^{-4} L^{-6}$, K/L rises 50%. Then y rises $(1.5)^{-4} - 1 \approx 17.6\%$, not 50%.

Source	Effect on Y/L	Diagram
$K/L \uparrow$	move along $f(k)$	deepening
$A \uparrow$	shift $f(k)$ up	TFP
$L \uparrow$, K fixed	dilution	
education	effective labour up	human cap

W10 SOLOW-SWAN

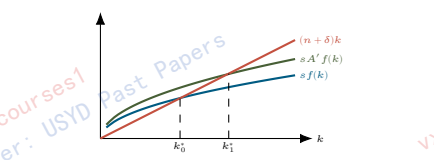
Solow law:
 $\Delta k = s A f(k) - (n + \delta) k$
Steady state where investment per worker equals break-even investment.

Consumption $c = (1 - s) y$.

Cobb-Douglas: if $y = A k^\alpha$, steady state satisfies $s A k^\alpha = (n + \delta) k$, so $k^* = (s A / (n + \delta))^{1/(1-\alpha)}$.

Saving-rate rise: $s f(k)$ shifts up; k^* , y^* rise; transition growth while $\Delta k > 0$; no permanent growth in y without TFP growth.

TFP rise: production and investment curves shift up; immediate y jump at old k plus gradual k accumulation to higher steady state.



Golden rule intuition: too little saving leaves low k ; too much saving sacrifices consumption. Max steady consumption where $MPK = n + \delta$ (if taught).

Convergence: poorer economies may grow faster if same A, s, n, δ ; conditional convergence fails with different institutions/policies.

Exam distinction: TFP is not capital deepening. TFP shifts function; deepening moves along it.

W11 EXCHANGE RATES

Nominal exchange rate: price of AUD in foreign currency (e.g. USD/AUD). Increase = AUD appreciation; decrease = depreciation. Cross rate via multiplication/inversion.

Real exchange rate:
 $q = e P^* / P$
when e is foreign currency per AUD. High q : Aus goods dearer, imports cheaper, $NX \downarrow$.

PPP: law of one price suggests $q \rightarrow 1$ LR; high domestic inflation tends to nominal depreciation. Fails SR due transport, nontraded goods, borders, pricing-to-market.

FX supply/demand: Australians supply AUD to buy imports/foreign assets; foreigners demand AUD to buy exports/Aus assets. Domestic interest rate up: demand AUD up, supply down, AUD appreciates.
Open-economy monetary channel: cash rate up appreciates AUD, lowers NX and import prices; helps inflation but hurts exporters. Floating rates absorb AD shocks.
Fixed rates: require reserve intervention; undervalued currency $\Rightarrow$ central bank buys foreign currency/sells domestic, reserves accumulate; overvalued drains reserves, attack risk.

Event	AUD market	e
Aus rates up	D up, S down	appreciates
Aus imports taste up	S up	depreciates
foreign boom	D up	appreciates
expected AUD fall	S up, D down	depreciates

PPP ex: if AU good ASD120, NZ good NZ\$125, PPP $e = 125/120 = 1.042$ NZD/AUD. Actual 1.12 means AUD strong/overvalued vs PPP.

W12 BOP OPEN ECONOMY

Balance of payments: records current, capital, financial transactions.

Ignoring errors,
 $CA + KA + FA = 0$
Current account includes trade balance, net income, transfers.

Financial account: net purchase/sale of assets. CA deficit must be financed by capital/financial inflow; not automatically bad if funds productive investment.

National saving identity: closed $S = I$; open
 $S = I + NCO$
and $NX = NCO$ in simple models. Govt deficit can reduce national saving and contribute to CA deficit: twin deficits.

Net exports: $NX = X - M$; affected by real exchange rate, domestic/foreign income, tastes, trade costs, supply capacity. J-curve: volumes adjust slowly after depreciation.

Net foreign assets: stock of external assets minus liabilities; persistent CA deficits reduce NFA, increasing future income outflows. Sustainability depends on returns/growth.

Policy evaluation: expansionary fiscal raises AD and imports, may raise rates/appreciate currency, worsening NX; supply-side reforms can raise capacity and competitiveness.

Account	Credit example	Debit example
CA	exports, income received	imports, income paid
KA	debt forgiveness received	capital transfer paid
FA	foreign buys Aus bond	Aus buys US shares

Mini ex: $CA = -80$, $KA = 15$. Then $FA = 65$ surplus so accounts sum zero. Interpret as net financial inflow financing CA deficit.

Final exam style: combine diagrams + arithmetic + policy judgement; always state direction, mechanism, and caveat.

CALCULATION TOOLKIT

GDP arithmetic: value added = sales - intermediate inputs; final-good method equals sum of VA. Real GDP with base b : $\sum p_b q_t$. Deflator isolates domestic output prices; CPI isolates consumer basket prices.

Approx growth rules: $\% \Delta (PY) \approx \pi + g_Y$; $\% \Delta (Y/L) \approx g_Y - g_L$; for Cobb-Douglas $y = A k^\alpha$, $\% \Delta y \approx \% \Delta A + \alpha \% \Delta k$.

Labour calculations: if participation and WAP known,
 $L F = \pi r t \times W A P$; if u known, $U = u \times L F$, $E = L F - U$. Do not include underemployed in U .

Inflation/interest: expected real rate uses expected inflation; realised redistribution uses actual inflation. If actual $\pi > \pi^*$, borrower repays less in real terms.

Multiplier workflow: identify autonomous shock ΔA ; compute MPE after leakages; $k = 1/(1 - MPE)$; sign from shock. Tax/transfer shocks first multiply disposable income by MPC.

Bank balance sheet: assets = reserves + securities + loans; liabilities = deposits/debt; equity = assets - liabilities. Insolvent if equity < 0 ; illiquid if cannot meet withdrawals despite positive equity.

Task	Formula	Check
CPI	basket current/base $\times 100$	fixed quantities
Deflator	$NGDP/RGDP \times 100$	domestic only
Output gap	$(Y - Y^*)/Y^*$	sign matters
BOP	$CA + KA + FA = 0$	solve residual
PPP	$e = P^*/P$	units!

Rounding: state formula first, keep 2-3 decimals in working, round final percent sensibly.

DIAGRAM PLAYBOOK

Keynesian cross: label Y and PAE ; 45-degree line; shock shifts PAE; horizontal gap at new intersection gives ΔY bigger than ΔA when multiplier > 1 .

AD-AS: label $Y, P, AD, SRAS, LRAS, Y^*$; demand shock moves AD; cost shock moves SRAS; productivity/capital moves LRAS (and often SRAS). Mention SR vs LR adjustment.

Solow: axes k and per-worker amounts; draw concave $s f(k)$ and straight $(n + \delta)k$; steady state at intersection; TFP shifts production/investment curve; saving shifts only investment curve.

FX market: vertical axis exchange rate (foreign currency per AUD if defined so); AUD appreciation is upward movement. Supply from Aus demand for foreign goods/assets; demand from foreign demand for Aus goods/assets.

Diagram	Must label	Movement
Cross	PAE , 45°, Y_0, Y_1	inventory logic
AD-AS	Y^*, Y_0, P_0	curve shift + eqm
Solow	$k^*, s f, (n + \delta)k$	transition arrows
FX	S, D, e	app/dep direction

Diagram marks usually require axes, curves, initial/final equilibrium, arrows, and one sentence mechanism.